

Erik-Jan Senn

PHD CANDIDATE · GRADUATE PROGRAMME IN ECONOMICS AND FINANCE

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Education

University of St. Gallen

St. Gallen, Switzerland

PHD IN ECONOMICS AND FINANCE

Mar 2022–present

- Research on modern data and machine learning for financial forecasting
- Coursework in econometrics, machine learning and finance, e.g. with Prof. A. Abadie, Prof. M. Cattaneo, Prof. V. Chernozhukov, Prof. P. Gagliardini, Prof. A. Krishnamurthy
- Supervisor: Prof. F. Audrino
- Research stay at **Aarhus University**, Denmark, Econometrics and Business Analytics section, hosted by B. Veliyev

University of St. Gallen

St. Gallen, Switzerland

MASTER OF ARTS IN QUANTITATIVE ECONOMICS AND FINANCE

Sep 2019–Feb 2022

- GPA: 5.7 (best in cohort)
- Thesis on realized covariance forecasting with neural networks
- Focus on financial econometrics, causal inference and machine learning
- Exchange at **Stockholm School of Economics**, Sweden

University of Tuebingen

Tuebingen, Germany

BACHELOR OF SCIENCE IN BUSINESS AND ECONOMICS

Oct 2015–Feb 2019

- GPA: 1.1 (best in cohort)
- Thesis: Measuring the Impact of MiFID II on Information Asymmetries Using Microstructure Models
- Focus on quantitative methods and finance
- Exchange at **San Jose State University**, United States

Professional Experience

University of St. Gallen (St. Gallen), Doctoral Research Assistant

Mar 2022–present

Financial forecasting, machine learning and teaching

(Python: NetworkX, Polars, scikit-learn, PyTorch, CVXPYlayers, Transformers; R)

University of St. Gallen (St. Gallen), Research Assistant

Dec 2019–Feb 2022

Market microstructure (Python) and experimental data analysis (R)

Deutsche Bundesbank (Frankfurt), Intern

Jun 2020–Aug 2020

Validation of a bond valuation tool for Eurosystem collateral (Python); interactive mispricing tool development (Excel/VBA)

Deutsche Bank AG (Frankfurt a. M.), Intern

Jun 2019–Aug 2019

FX portfolio backtesting for a large corporate client (Excel/VBA); derivatives pricing support (Bloomberg, Excel/VBA)

TOPSIM GmbH (Tuebingen), Developer and Working Student

Aug 2018–May 2019

Quantitative modelling of a retail chain simulation: prototyping (Excel/VBA) and product ownership for one development workstream (Jira)

Deloitte GmbH (Stuttgart), Working Student and Intern

Jan 2018–Jul 2018

Interactive tool development for client banks: loan pricing, prepayment prediction and interest rate risk analysis (R/Shiny)

Working and Conference Papers

Senn, E. *joint with* Phan, M. T. (2025). Read Them All at Once - Efficient Language Modeling for Very Long Financial Documents.

Senn, E. (2025). Model Recovery in Symbolic Regression: Theory, Conjectures and Open Questions.

Awards, Fellowships, & Grants

2026 **Mobility grant for doctoral students**, University of St. Gallen

26,000 CHF

2024 **Grant for doctoral writing retreat**, University of St. Gallen

2,500 CHF

Presentations

- Senn, E.** (2026). Financial Forecasting for Mean-variance Investors. Oral presentation: Aarhus Econometrics Lunch Seminar, Aarhus, Denmark.
- Senn, E.** (2025). A comparison of sequential and integrated estimation of realized volatility models used in portfolio optimization. Oral presentation: 19th International Conference on Computational and Financial Econometrics, London, United Kingdom.
- Senn, E.** *joint work with* Audrino, F.; Halousková, M.; Hirt, C.; Panzeri, G. (2025). Financial Social Networks. Poster: GPEF Day at University of St. Gallen 2025, St. Gallen, Switzerland.
- Senn, E.** (2025). Model Recovery in Symbolic Regression: Theory, Conjectures and Open Questions. Oral presentation: The Genetic and Evolutionary Computation Conference 2025 - Symbolic Regression Workshop, Málaga, Spain.
- Senn, E.** *joint work with* Audrino, F. (2024). The power of visuals: Using social media images for financial sentiment analysis. Oral presentation: 18th International Conference on Computational and Financial Econometrics, London, United Kingdom.
- Senn, E.** *joint work with* Phan, M. T. (2024). LongFinBERT: A Language Model for Long Financial Documents. Oral presentation: 1st Financial Fraud, Misconduct and Market Manipulation Conference, Lancaster, United Kingdom.
- Senn, E.** *joint work with* Phan, M. T. (2024). LongFinBERT: A Language Model for Long Financial Documents. Oral presentation: 26th International Conference of Computational Statistics, Giessen, Germany.
- Senn, E.** *joint work with* Phan, M. T. (2023). LongFinBERT: A Language Model for Long Financial Documents. Oral presentation: 17th International Conference on Computational and Financial Econometrics, Berlin, Germany.
- Senn, E.** *joint work with* Phan, M. T. (2023). LongFinBERT: A Language Model for Long Financial Documents. Poster: GPEF Day at University of St. Gallen 2023, St. Gallen, Switzerland.
- Senn, E.** (2023). Testing Financial Reference Dependence with Attention-based Reference Points. Oral presentation of research idea: Research Workshop, Gais, Switzerland.

Teaching Experience

2026	Agentic Workflows in Research and Teaching (all levels) , Lecturer (forthcoming)	<i>Hohenheim</i>
2025	Introduction to R (master's) , Lecturer	<i>St. Gallen</i>
2024	Understanding and Training Language Models (Master+) , Lecturer	<i>Hohenheim</i>
2023–present	Introduction to Programming in R (master's, 1 day) , Co-Lecturer	<i>St. Gallen</i>
2025–present	Data Science Fundamentals (bachelor's) , Teaching Assistant	<i>St. Gallen</i>
2024–present	Data Analytics II - Econometrics (bachelor's) , Teaching Assistant	<i>St. Gallen</i>
2023	Mathematics A (bachelor's) , Teaching Assistant	<i>St. Gallen</i>
2021–present	Data Analytics I - Statistics (bachelor's) , Teaching Assistant	<i>St. Gallen</i>
2016–2017	Introduction to Business and Economics (bachelor's) , Teaching Assistant	<i>Tuebingen</i>

Skills

Programming: Python, R [*advanced*]; SQL, SAS, VBA, GAUSS [*intermediate*]

Tools: Git, AI-assisted development [*advanced*]; Linux, MS Office, $\text{\LaTeX}$, Markdown, financial databases (Bloomberg, Reuters, TAQ, RavenPack)

Languages: German [*native*], English [*fluent*], French [*basic*]

Academic Service and Miscellaneous

- 2025–present **Peer review:** Neural Networks, Statistical Methods & Applications
- 2024–present **Linux server administrator for the Faculty of Mathematics and Statistics**
- 2024–present **Co-organiser of the St. Gallen Methods Reading Group**
- 2024–present **PhD programme representative**
- 2024–2026 **Member of the Tenure Track and Promotion Committee at the University of St. Gallen**
- 2023–2025 **Organiser of the internal Mathematics and Statistics Research Seminar**
- 2023 **Finalist in the Econometric Game, Amsterdam**
- 2022–2023 **Jury member for the Global Essay Competition of the St. Gallen Symposium**
- Lifelong **Bouldering, cycling and game enthusiast**